

# Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in $M \geq 6.5$ Earthquake Catalogs, 1973–2026 CE

© 2026 Yaroslav Marshalkin

---

Yaroslav Marshalkin

Corresponding author: marshalkin@gmail.com

Telegram: @MRSHLKN | GitHub: github.com/marshalkin-ux/paleoseismic-clustering

## Abstract.

*We analyse a merged catalog of 4,267 unique  $M \geq 6.5$  events (from 4,418 CSV rows; ~151 NOAA  $M < 6.5$  excluded from clustering) using the Baiesi–Paczuski metric  $\eta$  with tectonic-path distance (Bird 2003). The detector yields 47 algorithmic candidates (27 modern); however, ETAS validation ( $p_{\text{ETAS}}=1.0$ ) shows these candidates are indistinguishable from background activity. Primary result — negative (null/falsification): the detector finds on average 27.0 candidates in catalog-calibrated ETAS synthetic catalogs, matching  $N_{\text{obs}}=27$  — not specific to global series; reflects catalog background structure (see Sec. 5.5–5.6). Permutation  $p=0.0001$  (1/10,001) rejects temporal Poisson null only — not teleseismic triggering. Limitations — Sec. 5.5.*

Keywords: global seismicity; seismic series; earthquake clustering; heuristic metric with tectonic hint; Baiesi–Paczuski metric; ETAS validation; Monte Carlo; paleoseismology; remote triggering

---

## 1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake ( $M_w$  7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake ( $M_w$  9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that  $M \geq 7$  clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global  $M \geq 7$  and  $M \geq 8$  rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. Test (and if warranted, falsify) the hypothesis that physically meaningful multi-regional global series exist, with primary inference on the modern window (1973–2026), using complementary null tests (permutation vs ETAS) and explicit detector liberalness assessment. Scope. We combine nearest-neighbor clustering with heuristic tectonic hint and ETAS validation; this complements global rate tests (Michael 2011; Shearer & Stark 2012) with a different eta-linkage statistic but does not supersede their conclusions.

---

## 2. DATA AND METHODS

### 2.1 Catalog compilation

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records pre-1900, 47 events). Duplicates were merged using  $\pm 30$  days and  $\leq 50$  km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains 4,267 unique  $M \geq 6.5$  events (from 4,418 CSV rows;  $\sim 151$   $M < 6.5$  excluded from clustering). USGS ComCat: 2,088 raw (1973–2026)  $\rightarrow$  2,041 after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). `quality_score` is metadata, not an inclusion filter. Final catalog: 4,267  $M \geq 6.5$  events (4,418 CSV rows).

#### 2.1.1 Catalog completeness

Maximum-curvature analysis yields  $M_c = 6.55$ . Maximum-likelihood  $b$ -value from 1,688 events above  $M_c$ :

$$b = 0.911 \pm 0.018 \quad (n = 1,688 \text{ events})$$

### 2.2 Heuristic metric with tectonic hint

We tested an alternative to Euclidean distance (Bird 2003 graph). In 98% of pairs the implementation falls back to 1.5x great-circle distance — effectively scaled Euclidean; no improvement for global analysis (failed hypothesis test).  $r_{ij}$  is the shortest path between hypocenters along the Bird (2003) plate-boundary graph (20 key segments). Paths are computed with Dijkstra's algorithm (NetworkX). If either hypocenter lies >500 km from the nearest boundary node, or no graph path exists,  $r_{ij} = 1.5 \times r_{GC}$  (great-circle Haversine). Fallback audit (analyze\_tectonic\_fallback.py, fig07): 4987 pairs from 500 events; 98.0% GC fallback, 2.0% Dijkstra (4015 snap>500 km, 872 no path, 100 Dijkstra; 95 materially different; examples FE31 Japan, FE35 Philippines). Metric adds value for ~2% boundary-proximal pairs only. Tectonic diagnostic: median Deltalog10eta = +0.28.

$$\text{eta}_{ij} = t_{ij} * r_{ij}^{1.6} * 10^{(-b \cdot \text{mi})}$$

*b=1.0 per Baiesi & Paczuski (2004) for eta comparability; catalog b=0.911+/-0.018 in Monte Carlo null only.*

### 2.3 Analysis pipeline

Raw catalogs → dedup → exclude  $M < 6.5$  (~151 NOAA) → GK declustering (primary) → eta NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — different algorithms (GK: magnitude-dependent windows; ZBZ: eta NN + KDE threshold). At global  $M \geq 6.5$  scale ZBZ is permissive (sparse events → high eta). GK is sole primary for inference; ZBZ sensitivity only, does not replace GK. If ZBZ were primary,  $N=27$  likely unchanged (untested).

Why GK–ZBZ counts differ: not contradictory methods — GK removes short-range fore/aftershocks via fixed windows; ZBZ classifies only 1 event with exceptionally low eta at global catalog scale. Minimal ZBZ removal confirms declustering is not a critical fork.

### 2.4 Threshold $\text{eta}_0$ , series detection, and ETAS parameters

Automatic  $\text{eta}_0$  from nearest-neighbor eta distribution: KDE valley between bimodal  $\log_{10}(\text{eta})$  modes (Zaliapin & Ben-Zion 2013); default  $\text{eta}_0 = 10^{\text{median } \log_{10} \text{ eta}}$ . Validated against Poisson permutation null ( $n = 10,000$ ).

1. Declustering via Gardner–Knopoff [1974].
2. Nearest-neighbor forest: parent  $i^* = \text{argmin } \text{eta}_{ij}$  subject to  $\text{eta}_{ij} < \text{eta}_0$ .
3. Global series:  $N \geq 4$ ;  $M \geq 6.5$ ;  $\geq 3$  Flinn–Engdahl regions.
4. Sliding windows (1, 2, 5 yr); overlapping groups merged.

ETAS parameters (catalog-calibrated): MLE on 2,041 events (scripts/calibrate\_etas.py, results/etas\_calibration.json):  $\mu \approx 0.103$ ,  $K \approx 0.495$ ,  $\alpha \approx 0.063$ ,  $c \approx 10^{-4}$  d,  $p \approx 1.36$ . Literature defaults (Helmstetter & Sornette 2003) retained for comparison.

### 2.5 Statistical validation

Permutation test:  $n = 10,000$ ,  $p \leq 0.0001$ ,  $z = -6.17$  (modern). ETAS validation: calibrated params ( $\mu \approx 0.103$ ,  $K \approx 0.495$ ,  $\alpha \approx 0.063$ ,  $c \approx 10^{-4}$  d,  $p \approx 1.36$ ); 1000 catalogs/seed. Single seed=42: FPR=1000/1000; mean 27.0,  $p_{\text{ETAS}} = 1.0$ ;  $N_{\text{obs}}=27$ . Multiseed ETAS (seeds 42–51,  $n=1000$  each, results/etas\_multiseed.json): mean=27.0,  $\sigma=0.0$ , FPR=1.0,  $p_{\text{ETAS}}=1.0$  for all 10 seeds —

perfect stability because calibrated ETAS matches catalog event rate (~2001 background events). Literature  $\mu=0.008$  comparison only (earlier  $n=100$  runs):  $\text{mean} \approx 15.5$ ,  $p_{\text{ETAS}} \leq 0.001$ . Detector liberalism — see Sec.5.6 (FPR=1000/1000). Multiple comparisons (Methods): BH post-hoc on  $N=47$ , 45/47 at  $q=0.05$  — not discovery. Declustering: GK 2,017/2,041 (primary); ZBZ 2,040/2,041 (sensitivity only).

### 3. RESULTS

#### 3.1 Identified series

Full historical analysis yields 47 detector candidates (not validated global series): 27 modern ( $p \leq 0.0001$ ), 15 early instrumental ( $p = 0.007$ ; pre-1960 incompleteness caveat), 5 historical candidates (not significant,  $p = 0.46$ ; 47  $M \geq 6.5$  events pre-1900). 142 cluster candidates before filtering.

| ID        | N   | Regions | $M_{\text{max}}$ | Period    | Notes                                                                 |
|-----------|-----|---------|------------------|-----------|-----------------------------------------------------------------------|
| 1905–1910 | 193 | 43      | 8.8              | 1905–1910 | Largest early-instrumental candidate; catalog incomplete before ~1960 |
| S047      | 53  | 5       | 8.0              | 1982–2024 | Western Pacific subduction corridor                                   |
| S170      | 46  | 12      | 9.1              | 2002–2023 | Sunda belt; Sumatra 2004 (M 9.1)                                      |
| S095      | 25  | 4       | 7.9              | 1989–2017 | Western Pacific arc                                                   |
| S116      | 22  | 5       | 8.2              | 1993–2021 | South Pacific multi-arc series                                        |

Table 1. Top five multi-regional series ( $\geq 3$  Flinn–Engdahl regions).

#### 3.2 Spatial–temporal distribution

Elevated series activity occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic diagnostic: median  $\Delta \log_{10} \eta = +0.28$  on random pairs (mostly 1.5x GC fallback).

### 4. DISCUSSION AND CONCLUSIONS

The detector finds on average 27.0 candidates in catalog-calibrated ETAS synthetic catalogs, matching  $N_{\text{obs}}=27$  — not specific to global series; reflects catalog background structure. The global-series hypothesis is not supported (see Sec. 5.5). The permutation test rejects only a temporal Poisson null — trivial for earthquake catalogs with aftershocks.

1. Heuristic with tectonic hint: 98% GC fallback — failed hypothesis test.
2. 47 detector candidates indistinguishable from ETAS null; Delta CFS/dynamic stress — future work only.

Future work / supplement: Static Delta CFS and dynamic stress for S170, S047, S095; full ETAS MLE and multi-seed robustness.

### DATA AND CODE AVAILABILITY

Data and code: [github.com/marshalkin-ux/paleoseismic-clustering](https://github.com/marshalkin-ux/paleoseismic-clustering) (docs/data\_availability.md).  
External DOI deposition (Zenodo) deferred — GitHub only.

---

## REFERENCES

1. Baiesi M., Paczuski M. (2004). Scale-free networks of earthquakes and aftershocks. *Phys. Rev. E*, 69, 066106.
2. Zaliapin I. et al. (2008). Clustering analysis of seismicity and aftershock identification. *Phys. Rev. Lett.*, 101, 018501.
3. Zaliapin I., Ben-Zion Y. (2013). Earthquake clusters in southern California I. *J. Geophys. Res.*, 118, 2847–2864.
4. Bird P. (2003). An updated digital model of plate boundaries. *Geochem. Geophys. Geosyst.*, 4(3), 1027.
5. Hill D.P. et al. (1993). Seismicity remotely triggered by the magnitude 7.3 Landers earthquake. *Science*, 260, 1617–1623.
6. Brodsky E.E., Prejean S.G. (2006). New constraints on mechanisms of remotely triggered seismicity. *J. Geophys. Res.*, 111, B06312.
7. Pollitz F.F. et al. (1998). Viscosity of oceanic asthenosphere inferred from remote triggering. *Science*, 280, 1245–1249.
8. King G.C.P. et al. (1994). Static stress changes and the triggering of earthquakes. *BSSA*, 84, 935–953.
9. Stein R.S. (1999). The role of stress transfer in earthquake occurrence. *Nature*, 402, 605–609.
10. Michael A.J. (2011). Random variability explains apparent global clustering of large earthquakes. *Geophys. Res. Lett.*, 38, L21301.
11. Shearer P.M., Stark P.B. (2012). Global risk of big earthquakes has not recently increased. *PNAS*, 109(3), 717–721.
12. Ogata Y. (1988). Statistical models for earthquake occurrences and residual analysis. *J. Amer. Stat. Assoc.*, 83, 9–27.
13. Benjamini Y., Hochberg Y. (1995). Controlling the false discovery rate. *J. Roy. Stat. Soc. B*, 57(1), 289–300.
14. Gardner J.K., Knopoff L. (1974). Is the sequence of earthquakes in Southern California Poissonian? *BSSA*, 64, 1363–1367.
15. Woessner J., Wiemer S. (2005). Assessing the quality of earthquake catalogues. *BSSA*, 95, 684–698.
16. Young J.B. et al. (1996). The Flinn–Engdahl regionalization scheme: the 1995 revision. *Phys. Earth Planet. Int.*, 96, 223–297.

**Acknowledgments.** The author acknowledges USGS, NOAA NGDC, and the ISC for maintaining open seismic catalogs. This work received no targeted funding.